

TravelLink — UI/UX Design Plan

Mobile-First • Glassmorphism • Premium Experience

1. Codebase Analysis Summary

Backend Services (7 Controllers + 1 SignalR Hub)

Service	Controller	Key Endpoints	Status
Auth	AuthController	Login, Register, Refresh Token	✓ Complete
User	UserController	Profile, Search Users, Avatar Upload	✓ Complete
Friends	FriendController	Send/Accept/Reject Request, List, Unfriend	✓ Complete
Groups	GroupController	Create, List My Groups, Get By ID, Add/Remove Member, Leave, Cover Upload	✓ Complete
Trips	TripController	Create, List My Trips, Get By ID, Get By Group, Delete	✓ Complete
Expenses	ExpenseController	Create, Get By ID, Delete, Group Expenses, Balances, Analytics, Mark Paid, User-to-User Summary	✓ Complete
Itinerary	ItineraryController	Create Request, Add/Delete/Vote/Review Suggestions, Generate, Auto-Generate, Get Result, Toggle Item Complete, Trip Status	✓ Complete
Location	LocationHub (SignalR)	Join Group/Private Session, Send Location, Leave Session	✓ Complete

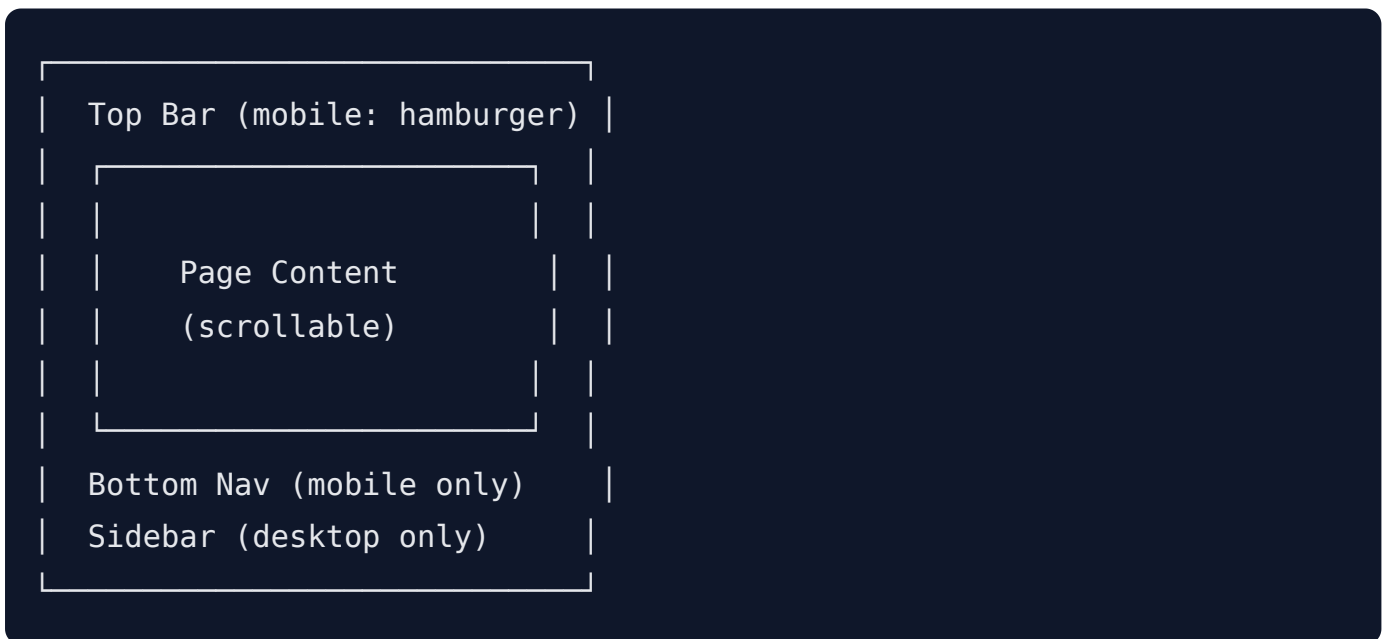
Frontend Status

Layer	Status
Types/DTOs	✓ All 8 type modules mapped
Services	✓ Auth, User, Group, Friend, Trip services
Auth Pages	✓ Login & Register with glassmorphism
Dashboard	✗ Empty placeholder
All Feature Pages	✗ Not built
Missing Services	✗ <code>expenseService.ts</code> , <code>itineraryService.ts</code> , <code>locationService.ts</code>

2. Service-Based Page Sections

Section A — App Shell (Layout)

The wrapper for all authenticated pages.



Components:

- `AppLayout.tsx` — Main shell with sidebar/bottom nav

- `TopBar.tsx` — Logo, search, theme toggle, notifications, avatar
 - `BottomNav.tsx` — Mobile: 5 tabs (Home, Trips, Friends, Map, Profile)
 - `Sidebar.tsx` — Desktop: Expanded navigation with sections
-

Section B — Dashboard (Home)

Route: `/dashboard`

Data Sources: `tripService.getMyTrips()`, `groupService.getMyGroups()`, `friendService.getPendingRequests()`

Layout:

- **Hero Card** — Greeting + next upcoming trip countdown
 - **Quick Actions Row** — Create Trip, Create Group, Add Friend, AI Planner
 - **Upcoming Trips Carousel** — Horizontal scroll of `TripDto` cards
 - **My Groups Grid** — 2-col grid of `GroupDto` cards with member avatars
 - **Pending Friend Requests** — Notification-style list
-

Section C — Trips (Core Feature)

Routes: `/trips`, `/trips/:tripId`

Data Sources: `tripService`, `groupService`, `itineraryService`

C1 — Trip List Page (`/trips`)

- Filter tabs: All | Upcoming | Past
- Trip cards showing destination, dates, group name, itinerary status
- FAB: "Create Trip" → opens bottom sheet / modal

C2 — Trip Detail Page (`/trips/:tripId`)

- Hero banner with destination image
- Tab navigation: **Overview** | **Itinerary** | **Expenses** | **Location**
- Overview: Trip info, group members, dates, countdown
- Links to sub-features scoped to this trip

C3 — Create Trip (Bottom Sheet / Modal)

- Form: Name, Destination, Dates (range picker), Select Group
 - Glass card form styling
-

Section D — Groups

Routes: `/groups`, `/groups/:groupId`

Data Sources: `groupService`, `tripService.getTripsByGroup()`

D1 — Groups List (`/groups`)

- Glass cards with cover image, name, member count, member avatar stack
- FAB: "Create Group"

D2 — Group Detail (`/groups/:groupId`)

- Cover hero image with glassmorphism overlay
- Member list with roles (Admin/Member)
- Trips by this group
- Actions: Add Member, Leave Group, Remove Member (admin)

D3 — Create Group (Modal)

- Form: Name, Description, Search & add friends as members
-

Section E — Expense Splitting

Routes: `/trips/:tripId/expenses`, `/expenses/user/:userId`

Data Sources: `expenseService` (needs to be created)

E1 — Group Expenses List

- Expense cards: Title, amount, paid by (avatar), date, category icon
- Split status indicators (settled/pending)
- FAB: "Add Expense"

E2 — Expense Detail

- Full breakdown with split amounts per member
- "Mark as Paid" button per split
- Animated progress bar for settlement

E3 — Group Balances View

- Who owes whom visualization
- `BalanceDto` → directional arrows between avatars
- Net balance per person (green = owed, red = owes)

E4 — Group Analytics

- Total spend, settled vs unsettled (donut chart)
- Category breakdown (horizontal bar chart)
- Member contributions comparison

E5 — User-to-User Summary

- Net balance with specific person
- Transaction history between two users
- Common groups

E6 — Add Expense (Full-page modal)

- Title, Amount, Category picker, Date
- Split Type selector: Equal | Exact | Percentage
- Participant selector from group members
- Split preview before submission

Section F — AI Itinerary Builder

Routes: `/trips/:tripId/itinerary`, `/trips/:tripId/itinerary/result`

Data Sources: `itineraryService` (needs to be created)

F1 — Itinerary Status Dashboard

- Status badge: Draft | Collecting Suggestions | Generated
- Budget overview, date range, suggestion progress
- Action buttons based on status

F2 — Suggestion Board

- Card list of `SuggestionDto` with vote counts
- Vote toggle (heart/thumbs up with animation)
- Admin review actions (approve/reject)
- "Add Suggestion" FAB

F3 — Auto-Generate Form

- Destination, dates, budget input
- Hotel config: cost per room, number of rooms
- Vehicle config: type, count, rental vs fuel costs
- Special notes textarea
- "Generate with AI ✨" CTA button

F4 — Itinerary Result View

- Budget summary header (total, hotel, vehicle, activities, remaining)
 - Day-by-day timeline with `ItineraryDayDto`
 - Items per day: place name, type icon, duration, cost, booking link
 - Checkbox to mark items complete
 - Dropped suggestions section with reasons
 - Per-person cost breakdown
-

Section G — Friends & Social

Routes: `/friends`

Data Sources: `friendService`, `userService.searchUsers()`

- **Tabs:** My Friends | Requests | Find People
 - Friend cards with avatar, name, email, action buttons
 - Pending requests with accept/reject actions
 - Search bar to find & add new friends
 - User-to-user expense summary link
-

Section H — Live Location Sharing

Routes: /trips/:tripId/location, /location/friend/:userId

Data Sources: LocationHub (SignalR)

- **Map View** — Full-screen map with member pins
 - Member avatars as map markers with live updates
 - Session controls: Join/Leave sharing
 - Group session vs Private (friend) session toggle
 - Accuracy & speed indicators per user
 - Connection status indicator
-

Section I — Profile & Settings

Routes: /profile

Data Sources: userService

- Avatar with upload overlay
 - Name, email display
 - Theme toggle (Dark/Light/System)
 - App preferences
 - Logout button
-

3. Design System — Glassmorphism Tokens

Color Palette

```

/* Dark Mode (Primary) */
--surface-base:      #0a0e1a;          /* Deep space background */
--surface-elevated:  rgba(255,255,255,0.03);
--glass-bg:          rgba(255,255,255,0.05);
--glass-border:      rgba(255,255,255,0.10);
--glass-highlight:   rgba(255,255,255,0.20);
--glass-blur:        20px;

/* Accent Palette */
--accent-sky:        #38bdf8;
--accent-emerald:    #10b981;
--accent-amber:      #f59e0b;
--accent-rose:       #f43f5e;
--accent-violet:     #8b5cf6;

/* Gradients */
--gradient-primary:  linear-gradient(135deg, #0ea5e9, #8b5cf6);
--gradient-success:  linear-gradient(135deg, #10b981, #38bdf8);
--gradient-warm:     linear-gradient(135deg, #f59e0b, #f43f5e);

```

Glass Card Variants

Variant	Background	Blur	Border	Use Case
glass-card	rgba(255,255,255,0.05)	20px	rgba(255,255,255,0.10)	Standard cards
glass-card-elevated	rgba(255,255,255,0.08)	24px	rgba(255,255,255,0.15)	Modals, sheets
glass-card-subtle	rgba(255,255,255,0.02)	12px	rgba(255,255,255,0.06)	Nested cards
glass-nav	rgba(10,14,26,0.80)	20px	Top border glow	Nav bars

glass - input	rgba(0,0,0,0.20)	8px	rgba(255,255,255,0.10)	Form inputs
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Typography

Headings:	'Playfair Display' – elegant serif for hero text
Body:	'Plus Jakarta Sans' – clean modern sans-serif
Brand:	'Great Vibes' – cursive for "Link" in logo
Monospace:	System monospace – for amounts/numbers

Micro-Animations

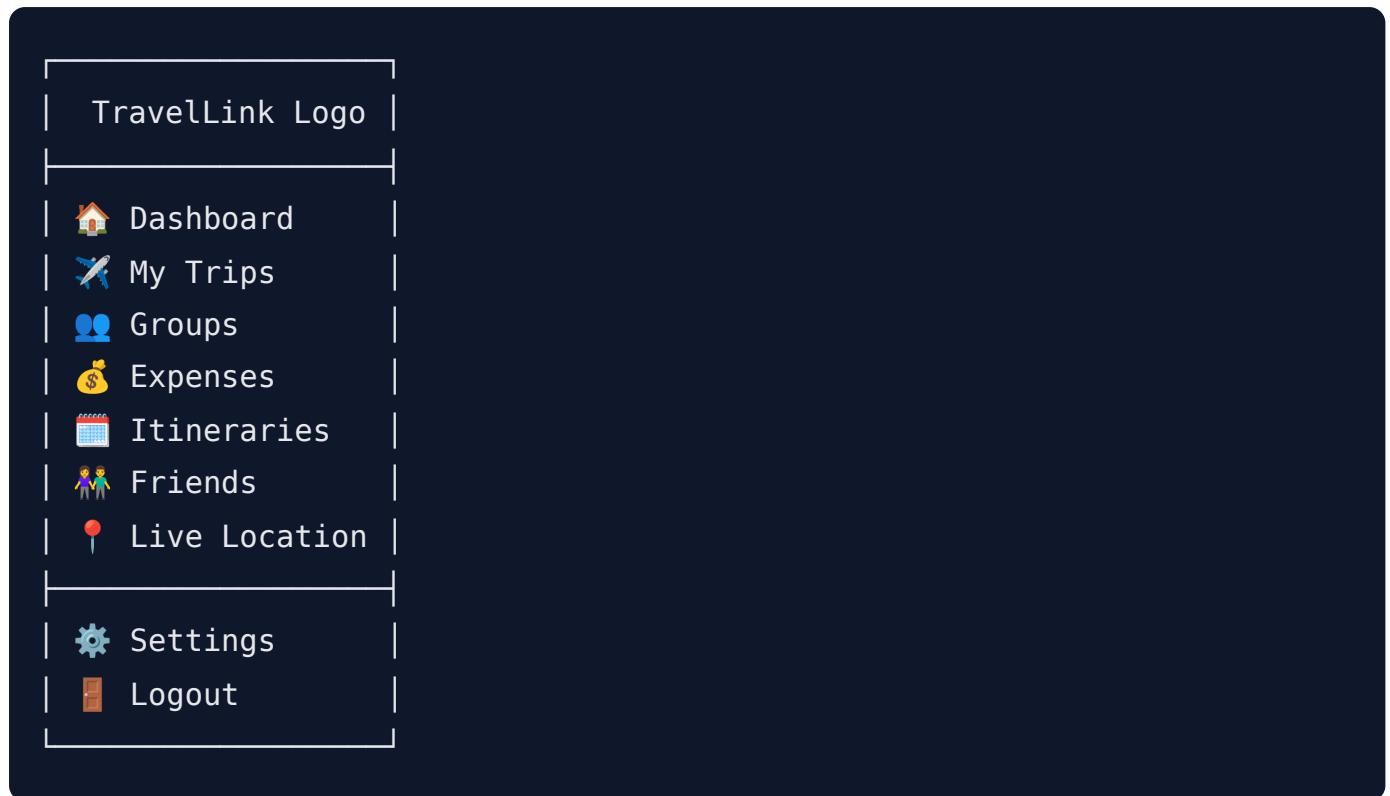
Element	Animation	Duration
Page enter	fadeInUp	400ms
Card hover	scale(1.02) + shadow lift	200ms
Button press	scale(0.97)	100ms
Tab switch	slideIn from direction	300ms
Vote toggle	pulse + scale bounce	300ms
Skeleton loading	shimmer gradient sweep	1.5s loop
Bottom sheet	slideUp from bottom	350ms ease-out
Toast	slideIn from right	300ms

4. Navigation Architecture

Mobile (Bottom Nav — 5 tabs)

				
Home	Trip	Frnd	Map	Prof

Desktop (Sidebar)



Route Map

<code>/login</code>	– LoginPage
<code>/register</code>	– RegisterPage
<code>/dashboard</code>	– Dashboard (Home)
<code>/trips</code>	– Trip List
<code>/trips/:tripId</code>	– Trip Detail (tabs: overview/itinerary/expenses/location)
<code>/groups</code>	– Group List
<code>/groups/:groupId</code>	– Group Detail
<code>/friends</code>	– Friends (tabs: list/requests/search)
<code>/profile</code>	– Profile & Settings

5. Missing Frontend Services to Create

`expenseService.ts`

```
// Endpoints needed:
addExpense(dto: CreateExpenseDto)
getExpenseById(id: string)
deleteExpense(id: string)
getGroupExpenses(groupId: string)
getGroupBalances(groupId: string)
getGroupAnalytics(groupId: string)
markSplitAsPaid(splitId: string)
getUserToUserSummary(userId: string)
```

itineraryService.ts

```
// Endpoints needed:
createRequest(dto: CreateItineraryRequestDto)
getTripStatus(tripId: string)
deleteRequest(requestId: string)
addSuggestion(requestId: string, dto: AddSuggestionDto)
deleteSuggestion(suggestionId: string)
toggleVote(suggestionId: string)
reviewSuggestion(suggestionId: string, dto: ReviewSuggestionDto)
getSuggestions(requestId: string)
generate(requestId: string, dto: GenerateItineraryDto)
autoGenerate(dto: AutoGenerateItineraryDto)
getResult(requestId: string)
toggleItemComplete(itemId: string)
```

locationService.ts (SignalR)

```
// SignalR Hub methods:
joinGroupSession(groupId: string)
joinPrivateSession(targetUserId: string)
sendLocation(location: LocationUpdateDto)
leaveSession()
// Listeners: LocationUpdate, UserJoined, UserLeft, Error
```

6. Component Tree

src/

```
├─ components/
│   ├─ layout/
│   │   ├─ AppLayout.tsx           - Main authenticated shell
│   │   ├─ TopBar.tsx             - Header with search & actions
│   │   ├─ Sidebar.tsx           - Desktop navigation
│   │   └─ BottomNav.tsx         - Mobile navigation
│   └─ ui/
│       ├─ GlassCard.tsx         - Reusable glass card wrapper
│       ├─ GlassModal.tsx        - Glass bottom sheet / modal
│       ├─ GlassInput.tsx        - Styled input component
│       ├─ Avatar.tsx            - User avatar with fallback
│       ├─ AvatarStack.tsx       - Overlapping avatar group
│       ├─ Badge.tsx             - Status/count badges
│       ├─ Button.tsx            - Primary/secondary/ghost variants
│       ├─ Skeleton.tsx         - Loading shimmer skeletons
│       ├─ EmptyState.tsx        - Empty data illustrations
│       ├─ FAB.tsx               - Floating action button
│       └─ TabBar.tsx            - Animated tab switcher
│   └─ ProtectedRoute.tsx
├─ features/
│   └─ auth/                     -  Done
│   └─ dashboard/
│       └─ DashboardPage.tsx
│   └─ trips/
│       ├─ TripsListPage.tsx
│       ├─ TripDetailPage.tsx
│       └─ CreateTripModal.tsx
│   └─ groups/
│       ├─ GroupsListPage.tsx
│       ├─ GroupDetailPage.tsx
│       └─ CreateGroupModal.tsx
│   └─ expenses/
│       ├─ ExpensesListPage.tsx
│       ├─ ExpenseDetailPage.tsx
│       ├─ BalancesView.tsx
│       └─ AnalyticsView.tsx
```

```

|   |   └─ AddExpenseModal.tsx
|   └─ itinerary/
|       └─ ItineraryStatusPage.tsx
|       └─ SuggestionBoard.tsx
|       └─ AutoGenerateForm.tsx
|       └─ ItineraryResultPage.tsx
|   └─ friends/
|       └─ FriendsPage.tsx
|   └─ location/
|       └─ LiveLocationPage.tsx
|   └─ profile/
|       └─ ProfilePage.tsx
└─ services/
    └─ api.ts                – ✓ Done
    └─ authService.ts       – ✓ Done
    └─ userService.ts       – ✓ Done
    └─ groupService.ts      – ✓ Done
    └─ friendService.ts     – ✓ Done
    └─ tripService.ts       – ✓ Done
    └─ expenseService.ts    – ✗ To build
    └─ itineraryService.ts  – ✗ To build
    └─ locationService.ts   – ✗ To build

```

7. Implementation Phases

Phase 1 — Foundation (App Shell + Design System)

1. Update `index.css` with complete glassmorphism design tokens
2. Build reusable UI components (`GlassCard` , `Button` , `Avatar` , `Skeleton` , etc.)
3. Build `AppLayout` with responsive `Sidebar` + `BottomNav` + `TopBar`
4. Create missing services (`expenseService` , `itineraryService` , `locationService`)

Phase 2 — Dashboard + Trips + Groups

5. `DashboardPage` — Hero, quick actions, upcoming trips, groups
6. `TripsListPage` + `TripDetailPage` + `CreateTripModal`

7. `GroupsListPage` + `GroupDetailPage` + `CreateGroupModal`

Phase 3 — Expenses + Friends

8. `ExpensesListPage` + `AddExpenseModal` + `ExpenseDetailPage`

9. `BalancesView` + `AnalyticsView`

10. `FriendsPage` with tabs

Phase 4 — AI Itinerary + Location

11. `ItineraryStatusPage` + `SuggestionBoard`

12. `AutoGenerateForm` + `ItineraryResultPage`

13. `LiveLocationPage` (SignalR integration)

Phase 5 — Profile + Polish

14. `ProfilePage` with avatar upload & settings

15. Animations, transitions, loading states

16. Error boundaries & empty states

8. Key Design Decisions

[!IMPORTANT]

- **Mobile-first:** All layouts designed for 375px width first, then scaled up
- **Glassmorphism everywhere:** Every card, modal, nav, and input uses frosted glass
- **Dark mode primary:** Deep space (`#0a0e1a`) base with luminous glass overlays
- **No placeholder images:** Use gradient backgrounds + icons for trip/group cards
- **Tailwind CSS v4:** Already configured, use `@theme` tokens for consistency
- **Bottom sheets over modals on mobile:** Create/Edit forms slide up from bottom
- **Skeleton loading:** Every data-dependent component shows shimmer while loading

[!TIP]

The existing auth pages already establish the glassmorphism pattern with `glass-card`, `input-field`, `btn-primary`, `bg-mesh`, `text-nature-gradient`, and `animate-fade-in-up` — all new pages should extend these patterns.